Reinforcement-Learning-Guided Latent Diffusion for Visual Counterfactual Explanations in Chest Radiography

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Abstract

Counterfactual Validity (CV) — the fraction of generated counterfactuals that flip a classifier's decision — is the standard headline metric for visual counterfactual generation, but when CV is scored with the same classifier (or classifier family) used to *guide* generation, a system can improve CV by exploiting that classifier's blind spots rather than by producing a genuinely better counterfactual, and CV alone cannot tell the two apart. This report's main contribution is a system-and-evaluation pair that makes this failure mode visible and partially closes it: a Proximal Policy Optimization (PPO) agent that *controls* a LoRA-fine-tuned Stable Diffusion sampling process step-by-step (per-step noise strength, classifier- and preservation-guidance scales, refinement passes, stopping decision) under a segmentation-mask-gated background lock, evaluated not just with CV but with a lesion-annotation-anchored Lesion Hit Rate (LHR) metric that does not depend on the guidance classifier at all, plus a held-out-transfer protocol scored on classifier backbones excluded from guidance. Run across five trained arms (no countermeasure, single- and multi-model ensemble guidance, a Grad-CAM spatial constraint, and combinations), the two evaluation families diverge sharply: guidance-ensemble countermeasures that raise LHR lift and held-out transfer *lower* mean CV (to 0.838) relative to the no-countermeasure baseline, exactly the direction a CV-only evaluation would read as a regression. Two further arms produce confirmed, not merely unproven, negative results under paired testing — a sparsity-reward retrain (Cohen's $d = 0.512$) and a Grad-CAM-constrained retrain (Cohen's $d = 1.205$, the largest paired effect size measured anywhere in this project) are both significantly worse than the 2-model-ensemble default — which we read as evidence that the LHR/transfer protocol has discriminative power CV lacks, not as a weakness of the underlying system. A separate audit of the evaluation apparatus itself (patient-level splitting, seeding, control and countermeasure arms, and the held-out-transfer machinery) found and closed a routing bug and a denominator bug that would each have silently invalidated an arm comparison, and corrected a set of hand-altered values discovered in the recorded results (Section 8). On the RSNA Pneumonia Detection Challenge dataset, following Singla *et al.* [1]'s protocol, the system separately achieves a mean CV of 0.940 (n→p: 0.980, p→n: 0.900 at $\tau = 0.8$), an XRV-DenseNet FID of 20.02 / 5.63 against each direction's real target class, and, on two clinical proxy metrics (Cardio-Thoracic Ratio, Costophrenic-recess score), no significant shift in this evaluation run — CTR's within-image delta is not distinguishable from zero ($p = 0.69$/ 0.44) and SCP's own computation flags itself as degenerate. We report the full evaluation suite, analyze a persistent asymmetry between the two edit directions, and discuss the reward-shaping and background-locking design decisions that resolved earlier failure modes (diffuse whole-image edits, degenerate reward scaling). Appendix A reports exploratory, not-yet-validated work extending the same pipeline to a second, dissimilar imaging domain (mammography, CBIS-DDSM).

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1 Introduction

Deep classifiers deployed on chest radiographs (CXR) are typically audited with attribution maps (Grad-CAM, integrated gradients), which highlight *where* evidence is concentrated but say nothing about *what change* would alter the prediction. **Visual counterfactual explanations** close this gap: for a query image x classified $f(x)$, a counterfactual generator produces x_c that (i) a downstream classifier scores at the target class c, (ii) remains visually realistic, and (iii) differs from x only in the minimal, class-relevant region — preserving patient anatomy, foreign objects (tubes, pacemakers, monitoring leads), and overall image identity.

This project builds such a system for binary pneumonia classification on the RSNA Pneumonia Detection Challenge dataset, combining three ingredients:

1. A **LoRA-fine-tuned Stable Diffusion v1.5** latent generator, adapted from natural images to chest radiographs via a lightweight denoising fine-tune of its cross-attention layers only.

2. A **gradient-guided DDIM sampling loop** that steers the reverse diffusion trajectory toward the target class using classifier gradients, while a segmentation-mask-gated preservation term and a RePaint/Blended-Latent-Diffusion-style background lock keep edits confined to the lung field.

3. A **PPO agent** that treats the per-step guidance hyperparameters (noise strength, classifier-guidance scale, preservation-guidance scale, number of refinement passes, stopping decision, guidance ramp-up point) as a *continuous action space*, learning a policy that adaptively schedules these controls instead of using one fixed schedule for every image.

The central hypothesis is that a fixed, hand-tuned guidance schedule is a poor fit across a heterogeneous test population: some images need only a gentle nudge to cross the decision boundary, others require aggressive, multi-pass editing, and a fixed schedule either under-edits the latter or over-edits (and de-realizes) the former. By framing per-step guidance control as a sequential decision problem and training it with reward signal that explicitly trades off classification progress against preservation and realism, the agent can learn an image-adaptive editing policy.

A second, equally central concern of this report is that the standard way of *evaluating* such a system is weaker than it looks. Counterfactual Validity is scored by re-running a classifier on the generated image, and when that scoring classifier is drawn from the same family as the classifier used to *guide* generation, a countermeasure that pushes CV up can be doing so by exploiting a shared blind spot rather than by producing a more genuine edit — CV alone cannot distinguish the two. Section 8 makes this concrete with a five-arm ablation: countermeasures that raise a lesion-annotation-anchored Lesion Hit Rate (LHR, independent of the guidance classifier) and a held-out transfer ratio (scored on classifier backbones excluded from guidance) can *lower* mean CV relative to a no-countermeasure baseline — the opposite of what a CV-only read would conclude — and two further arms produce confirmed, large-effect-size negative results under paired testing, which we treat as evidence the LHR/transfer protocol has discriminative power CV lacks, not as a weakness of the underlying system.

We evaluate the RSNA chest-radiography system with the metric suite of Singla *et al.* [1] — Counterfactual Validity (CV), Fréchet Inception Distance (FID), and Foreign-Object Preservation (FOP) — plus a model-free Identity Preservation (IP) score introduced here as a practical substitute for FOP, two clinically motivated proxy metrics, Cardio-Thoracic Ratio (CTR) and Costophrenic-recess score (SCP), each tested with paired t-tests between real and counterfactual populations, and, in Section 8, the classifier-independent LHR and held-out-transfer protocol that this report treats as its primary evaluation contribution. RSNA chest radiography is the focus of this report and the domain on which every design decision above was made and validated; Appendix A reports, as exploratory work kept out of the main body for the reasons given there, a preliminary extension of the same pipeline to a second, unrelated imaging domain.

2 Related Work

Counterfactual explanations in medical imaging. Singla *et al.* [1] introduced a GAN-based counterfactual generator for chest X-rays with an explicit evaluation protocol — CV, FID, and FOP — that we adopt here as the benchmark for comparability. Their FOP metric relies on a custom Fast R-CNN detector for foreign objects trained on private, manually annotated data, which is not reproducible offline; we discuss our substitute (Section 4.5.3) below.

Diffusion-based counterfactuals. Diffusion Autoencoders and classifier/gradient-guided diffusion sampling (e.g., DiffAE, guided diffusion, SDEdit) have become the dominant paradigm for counterfactual and editable image generation, replacing earlier GAN-based approaches due to their more stable training and superior sample fidelity. This project adopts a **gradient-guided DDIM** scheme built on a pretrained, LoRA-adapted Stable Diffusion v1.5 backbone rather than training a diffusion model from scratch on the (comparatively small) CXR dataset.

RePaint / Blended Latent Diffusion. Spatially-constrained inpainting methods that repeatedly renoise and re-composite the unedited region at every denoising step [2, 3] are adapted here as a hard *background lock*: instead of trusting the guidance gradient alone to keep edits inside the lung field, the region outside a dilated, feathered lung-segmentation mask is forcibly reset to the original image's forward-diffused latent at every step.

RL for controllable diffusion. Framing the choice of diffusion sampling hyperparameters (guidance scale, number of steps, noise schedule) as an RL control problem is comparatively less explored than RLHF-style fine-tuning of the diffusion *weights* (e.g., DDPO, reward-weighted fine-tuning). Here PPO does not touch the diffusion model parameters at all; it controls only the sampling procedure around a frozen (LoRA-fine-tuned) generator, which keeps the action space small, interpretable, and reusable across arbitrary query images at inference time.

3 Method

3.1 Problem formulation

Given a frozen binary classifier $f : \mathcal{X} \to [0, 1]$ (pneumonia probability) and a target class $c \in \{0, 1\}$, the goal is to find x_c minimizing

$$\mathcal{L}(x_c) \;=\; \underbrace{\mathrm{BCE}\big(f(x_c),\, c\big)}_{\text{classification}} \;+\; \underbrace{\lambda_{\text{pres}}\, d(x, x_c)}_{\text{identity preservation}}$$

where d is a masked distance that penalizes change *inside* the segmented lung field and change to a frozen object-detector embedding (foreign-object proxy), while leaving change *outside* the lung field essentially free (subject to the hard background lock described below).

3.2 Latent diffusion generator

The generator (`src/generator.py`) wraps a pretrained SD1.5 VAE and UNet. Images are encoded to a $4 \times 32 \times 32$ latent via the frozen VAE encoder; a lightweight `cond_proj` MLP maps (c, pooled latent) to a single cross-attention conditioning token consumed by the UNet — this replaces SD1.5's native text conditioning with a scalar class-conditioning signal appropriate for a binary medical classification task.

LoRA fine-tuning. Rank-8 LoRA adapters ($\alpha = 16$, dropout 0.05) are inserted into the UNet's `to_q`, `to_k`, `to_v`, `to_out.0` projections and trained (together with `cond_proj`, upcast to fp32) on the standard ϵ-prediction denoising loss over the RSNA training set, adapting the natural-image prior toward the chest-radiograph domain without touching the frozen VAE or the bulk of the UNet weights.

Gradient-guided DDIM sampling. Counterfactual generation runs partial DDIM sampling (an SDEdit-style forward-then-reverse trajectory whose length is set by a `strength` fraction of the full

$T = 50$ steps) with two gradient guidance terms injected into the predicted $\hat{z}_0$ at every step:

$$\Delta z_0 \;=\; -\,w(i)\,\alpha_{\text{cls}}\,\nabla_{\hat{z}_0}\mathcal{L}_{\text{cls}} \;-\; w(i)\,\alpha_{\text{pres}}\,\nabla_{\hat{z}_0}\big(\underbrace{\text{seg-weighted L1}}_{\text{anatomy}} + \eta\,\underbrace{\text{obj-det MSE}}_{\text{foreign objects}}\big)$$

gated by a mask M (dilated + Gaussian-feathered lung segmentation) and a linear ramp $w(i)$ that rises from 0 to 1 starting at a configurable fraction of the trajectory, avoiding the discontinuity of an earlier hard on/off guidance switch.

Background lock. At every step, the latent *outside* the edit mask is overwritten with the original image's latent, forward-noised to the current timestep (clean at the final step) — a RePaint / Blended Latent Diffusion composite. This is a hard spatial constraint, independent of how well the gradient guidance happens to behave: content outside the lung field is guaranteed to collapse to the VAE reconstruction floor rather than being repainted by the free SDEdit process, which is what an unconstrained, high-`strength` schedule would otherwise do to the entire thorax.

3.3 PPO-controlled sampling schedule

Rather than fixing `strength`, guidance scales, and pass count for every image, a PPO agent (`factory/ppo.py`, `factory/environment.py`) selects a 9-dimensional continuous action at each of up to $T = 25$ refinement steps, mapped through `sigmoid`-based transforms to:

Action	Controls	Range
a_0	DDIM strength	$[0.20,\ 0.70]$
a_1	classifier-guidance scale	$[0.5,\ 3.0]$
a_2	preservation-guidance scale	$[0.3,\ 1.5]$
a_3	stop signal τ_t	$[0,\ 1]$
a_4, a_5	$+1$ extra refinement pass each (Bernoulli gates)	$\{0, 1\}$
a_6	per-pass strength boost	$[0.85,\ 1.15]$
a_7	guidance bias (cls↑ vs. pres↓ trade-off)	$[0.70,\ 1.30]$
a_8	guidance ramp-up start fraction	$[0.1,\ 0.5]$

The environment's observation concatenates the flattened original and current latents, normalized step index, current $|f(x_t) - c|$, the target c, and the flattened lung mask. Episodes terminate when $|f(x_t) - c| < \varepsilon_{\text{stop}} = 0.08$ or the step budget is exhausted.

Reward. Each step's shaped reward combines progress toward the target (R_{prog}, R_{val}, R_{close2}), a fidelity penalty against the original image (R_{fid}), a mask-weighted preservation penalty over the lung region plus a foreign-object-embedding KL term (R_{pres}), a per-step budget cost (R_{bud}), and terminal bonuses for reaching validity (R_{hit}) and for stopping once valid (R_{stop}), each weighted by the λ coefficients in Table 1.

Actor-critic architecture and update. A shared 512→256 MLP trunk feeds a Gaussian policy head (learnable, per-dimension log-std), a value head, and three auxiliary heads (feasibility, progress, stop) that are trained with auxiliary losses alongside the standard clipped PPO objective and GAE advantage estimation ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$). A τ-gated shaping term blends the raw step reward with a scaled advantage (`DELTA_MIX`$= 0.2$) and is used as the value-regression target, giving the auxiliary progress/stop signal a channel into the value function beyond the raw GAE bootstrap.

3.4 Two directional policies

`MedicalCounterfactualEnv` is built around a single fixed c_{target}, so one policy is trained for normal→pneumonia ($c = 1$) and a separate one for pneumonia→normal ($c = 0$), sharing the same fine-tuned generator but specializing their sampling schedules. This is motivated by an observed asymmetry: n→p is empirically the harder direction (lower pre-fix CV, more diffuse edits), plausibly because

"adding" a plausible opacity pattern to a clean lung field is a harder generative problem than "removing" one, and because the shared reward's `direction_scale` historically applied unevenly across reward terms (Section 6.1).

4 Experimental Setup

4.1 Dataset

The RSNA Pneumonia Detection Challenge dataset (Kaggle) supplies chest radiographs (DICOM, min–max normalized to $[0, 1]$, resized to 256×256) with a binary label collapsed from bounding-box annotations: an image is labeled positive if any bounding box marks lung opacity (pneumonia), negative otherwise.

4.2 Pretrained components

- **Classifier** f: TorchXRayVision DenseNet-121 (`densenet121-res224-rsna`), pretrained directly on the RSNA pneumonia task — the same label space as this project's target.
- **Segmentation net** S: TorchXRayVision PSPNet, providing lung and heart channel masks used both for the preservation reward and the CTR clinical metric.
- **Object detector** O: YOLOv8n feature extractor, used as a coarse foreign-object / structural-change embedding for the preservation KL term and the SCP proxy metric.

4.3 Hyperparameters

Table 1: Key configuration values (`config.py`).

Group	Parameter	Value
Image / latent	Image size / latent size	256 / 32
Diffusion	DDIM steps (T_{diff})	50
Diffusion guidance	α_{cls} / α_{pres}	4.0 / 0.05
Background lock	mask dilation / feather σ	2 / 1.0 (latent cells)
PPO environment	budget T / $\varepsilon_{\text{stop}}$	25 / 0.08
PPO reward weights	$\lambda_{\text{prog}}, \lambda_{\text{val}}, \lambda_{\text{fid}}, \lambda_{\text{pres}}$	1.5, 2.0, 0.5, 0.5
	$\lambda_{\text{bud}}, \lambda_{\text{stop}}, \lambda_{\text{hit}}, \lambda_{\text{close2}}$	0.1, 8.0, 6.0, 2.5
GAE	γ / λ_{GAE} / DELTA_MIX	0.99 / 0.95 / 0.2
PPO update	clip / value coef / entropy coef	0.2 / 0.5 / 0.01
LoRA	rank / α / dropout	8 / 16 / 0.05
Optimization	learning rate (PPO & LoRA)	3×10^{-4} / 1×10^{-4}

Training and evaluation were run on a single Tesla V100-32GB node. `N_EPISODES` and `N_DENOISE_EPOCHS` were set as compute-budget placeholders pending a short throughput benchmark on this hardware, per the guidance recorded in `config.py`.

4.4 Evaluation protocol

Following [1], the evaluation set is split by classifier confidence into a negative group $X_n = \{x : f(x) < 0.2\}$ and a positive group $X_p = \{x : f(x) > 0.8\}$ (up to $n_{\text{eval}} = 200$ each). The trained n→p policy generates $X_c^{n\to p}$ from X_n (target $c = 1$) and the trained p→n policy generates $X_c^{p\to n}$ from X_p (target $c = 0$), using the policy's deterministic (mean) action for reproducibility.

4.5 Evaluation metrics

4.5.1 Counterfactual Validity (CV)

$$\mathrm{CV} = \Pr\left[\, |f(x_c) - f(x)| > \tau \;\wedge\; (f(x_c) - 0.5) \text{ has the intended sign} \,\right], \quad \tau = 0.8$$

A counterfactual only counts as valid if it both moves the classifier score by a large margin *and* crosses the decision boundary in the direction demanded by the target — a direction-aware tightening of the original direction-agnostic formulation, which would otherwise silently credit an image that moved the wrong way by a large margin.

4.5.2 Fréchet Inception Distance (FID)

Distributional realism is measured between the real group and its counterfactual set using penultimate-layer features. We use a TorchXRayVision DenseNet-121 (`-rsna` weights) feature extractor rather than the classical ImageNet Inception-v3 backbone, since it is trained on the same source distribution (RSNA) as the evaluation data, giving a feature space directly sensitive to pneumonia-relevant visual structure rather than generic natural-image texture. Covariance matrices are regularized ($\epsilon = 10^{-6}$) before the matrix square root to avoid the numerical instability that arises when sample count is smaller than feature dimensionality.

4.5.3 Identity Preservation (IP) — a model-free FOP substitute

The paper's Foreign-Object Preservation (FOP) metric requires a custom-trained foreign-object detector (a Fast R-CNN trained on private, manually annotated data) that is not reproducible offline; no TorchXRayVision classifier exposes a usable "Support Devices" head under inspection. We report IP as a practical, dependency-free substitute measuring the same underlying property — whether patient-specific, non-pathology content survives editing:

- **MS-SSIM**: multi-scale structural similarity between x and x_c, capturing global anatomical identity across scales.
- **HF-corr**: Pearson correlation of high-pass-filtered (image minus Gaussian blur) versions of x and x_c, a direct proxy for whether thin, high-contrast structures — tubes, leads, pacemakers — survive editing, since these are exactly the spatial-frequency content a Gaussian high-pass isolates.

4.5.4 Clinical proxy metrics: CTR and SCP

Cardio-Thoracic Ratio (CTR) is the ratio of the heart mask's horizontal span to the lung mask's horizontal span (both from the frozen PSPNet segmenter); CTR > 0.5 is the standard clinical cardiomegaly threshold. **Costophrenic-recess score (SCP)** is a proxy for pleural-effusion-related blunting, approximated here via a sigmoid transform of the YOLOv8 feature-norm in the lower-lobe crop (low norm $\rightarrow$ high SCP $\rightarrow$ healthy/un-blunted recess). Both are reported as group means $\pm$ std for $X_n, X_p, X_c^{n\to p}, X_c^{p\to n}$, together with a paired t-test on the within-image delta $X - X_c$.

5 Results

5.1 Quantitative summary

Table 2 reports the full metric suite computed by `run_full_evaluation` (values from `checkpoints/eval_metrics.json`).

Neither clinical proxy metric reaches significance in the current evaluation run. CTR's within-image delta is not statistically distinguishable from zero in either direction ($p = 0.69$ for n$\rightarrow$p, $p = 0.44$ for p$\rightarrow$n) — consistent with the background lock holding the cardiac silhouette essentially untouched,

Table 2: Evaluation results, PPO-guided generation, $n_{\text{eval}} \leq 200$ per group.

Metric	Definition	n→p	p→n
CV	fraction valid ($\tau = 0.8$)	0.980	0.900
	mean	0.940	
FID	XRV-DenseNet features, vs. real target class, lower = better	20.02	5.63
IP	MS-SSIM	0.920	0.880
IP	HF-corr	0.949	0.919

Table 3: Clinical proxy metrics: group means ± std and paired t-tests on the real → counterfactual delta.

	X_n	$X_c^{n\to p}$	$\Delta\ (t,\ p)$
CTR	0.417 ± 0.054	0.417 ± 0.057	$t = 0.40,\ p = 0.69$
SCP	0.479 ± 0.001	0.480 ± 0.001	degenerate[‡]
	X_p	$X_c^{p\to n}$	$\Delta\ (t,\ p)$
CTR	0.463 ± 0.107	0.534 ± 1.104	$t = -0.77,\ p = 0.44$
SCP	0.479 ± 0.001	0.480 ± 0.001	degenerate[‡]

[‡]SCP's four group means span a relative spread of 0.21% (`checkpoints/eval_metrics.json`); the source metrics code itself flags this as degenerate and withholds the delta/t-test, noting they "were computed on a constant and must not be reported." Group means are retained above for completeness; no SCP shift claim is made.

but this run does not support a claim of a *detected*, significant CTR shift, only that no CTR effect was measurable at $n \approx 200$. SCP's own computation flags itself as degenerate in this run (relative spread 0.21% across all four groups, i.e. effectively constant), so its delta and t-test are withheld rather than reported — the previously stated "SCP drops, directionally consistent with expected blunting" finding does not hold in this run's data and has been retracted rather than reinterpreted. This is a weaker clinical-proxy result than earlier drafts of this report claimed; it does not by itself indicate the counterfactuals are clinically inert, since CTR and SCP are coarse proxies for a narrow pair of findings (cardiomegaly, pleural effusion) that need not respond to every edit, but it means this evaluation run offers no statistical evidence either way, and it should be re-run and reconciled with Table 2's n_{eval} before either metric is asserted again.

5.2 Comparison with baseline methods

Table 4 places PPO-CF alongside two baselines, discussed further in Section 6.4: a **Vanilla CycleGAN** (an unconditional, unguided image-to-image baseline with no classifier or preservation guidance) and **GANterfactual**, the conditional GAN-based counterfactual generator of Singla *et al.* [1]. PPO-CF trails GANterfactual slightly on mean Counterfactual Validity (mean CV 0.940 vs. 0.9725) while producing a substantially more realistic distribution of generated images: FID (against each direction's real target class) of 20.02 vs. 25.0 for n→p ($1.25\times$ better) and 5.63 vs. 20.0 for p→n ($3.6\times$ better) — a real but direction-dependent gap, not the uniform "4–6×" previously claimed. PPO-CF also improves HF-corr identity preservation in both directions (0.95 n→p / 0.92 p→n vs. GANterfactual's 0.75 / 0.73). The unguided Vanilla CycleGAN baseline, lacking any classifier-guidance or background-locking mechanism, validates far less reliably (mean CV 0.385) and produces markedly less realistic images (FID 55.0/45.0), confirming that gradient guidance and the background lock — not merely the diffusion backbone itself — are responsible for the bulk of the validity and realism gains.

Table 5 extends the comparison to the two clinical proxy metrics, reporting the mean within-image delta and its paired t-test significance for each direction. PPO-CF's CTR shift is not statistically significant in this evaluation run ($p = 0.69$ n→p, $p = 0.44$ p→n) — unlike the two baselines, which both shift

Table 4: Comparison with baseline counterfactual generation methods. CV and FID are computed as in Sections 5.2 and above; IP is reported as MS-SSIM / HF-corr. Best value per column in **bold**. PPO-CF's CV, FID, and IP columns are read directly from `checkpoints/eval_metrics.json` (CV mean $= (0.980 + 0.900)/2 = 0.940$; FID from the `n_to_p_vs_real_positive` / `p_to_n_vs_real_negative` fields — the cross-group FID against each direction's real target class, i.e. the realism test proper, as opposed to the same-group `negative_group`/`positive_group` fields). †Mean CV of 0.940 is below GANterfactual's printed 0.947; the bold mark moves accordingly.

	CV ↑		FID ↓		IP (MS-SSIM / HF-corr) ↑		
Method	n→p	p→n	n→p	p→n	n→p	p→n	**Mean CV**
Vanilla CycleGAN	0.424	0.346	55.0	45.0	0.55 / 0.58	0.52 / 0.56	0.385
GANterfactual [1]	**0.998**	0.872	25.0	20.0	0.72 / 0.75	0.74 / 0.73	**0.947**
PPO-CF (ours)	0.980	**0.900**	**20.02**	**5.63**	**0.92 / 0.95**	**0.88 / 0.92**	0.940†

CTR significantly. This is consistent with the background lock's design goal of confining edits to the lung field and leaving the cardiac silhouette essentially untouched, but it means PPO-CF's CTR result here is an absence of a detected effect, not a small-but-significant one as earlier drafts of this report stated. PPO-CF's SCP metric is flagged degenerate in this run (Table 3) and no delta is reported for it; both baselines' SCP shifts are taken as given (Section 6.4).

Table 5: Comparison of clinical proxy-metric shifts across methods: mean paired delta (t, p) by direction. PPO-CF's row is measured from this project's own per-image data (Table 3); the Vanilla CycleGAN and GANterfactual rows are taken as given (Section 6.4).

	CTR Δ (t, p)		SCP Δ (t, p)	
Method	n→p	p→n	n→p	p→n
Vanilla CycleGAN	0.030 ($t = 2.0$, $p = 0.05$)	0.040 ($t = 2.3$, $p = 0.02$)	-0.0060 ($t = -8.0$, $p = 1\times10^{-7}$)	-0.0070 ($t = -8.5$, $p = 1\times10^{-8}$)
GANterfactual	0.070 ($t = 4.2$, $p = 3\times10^{-5}$)	0.090 ($t = 5.5$, $p = 5\times10^{-7}$)	-0.0030 ($t = -6.5$, $p = 1\times10^{-6}$)	-0.0035 ($t = -7.0$, $p = 5\times10^{-7}$)
PPO-CF (ours)	$+0.0014$ ($t = 0.40$, $p = 0.69$)	-0.0643 ($t = -0.77$, $p = 0.44$)	degenerate‡	degenerate‡

‡See Table 3's footnote — PPO-CF's SCP delta/t-test are withheld as degenerate in this run, not zero or absent by design.

5.3 Qualitative counterfactuals

Figures 1 and 2 show representative examples for each direction, generated with the deterministic (mean-action) policy on a confidently-negative and confidently-positive query image respectively (the same selection criterion, $f(x) < 0.2$ / $f(x) > 0.8$, used for the quantitative X_n/X_p groups).

In both examples the classifier score fully saturates toward the target ($f(x_c) < 0.01$ or > 0.99), well beyond the CV validity margin ($\tau = 0.8$), and gross anatomical layout (rib cage, mediastinal silhouette, external hardware visible in the p→n example) is visibly retained outside the edited lung region — the qualitative signature of the background-lock constraint holding as intended.

5.4 Training diagnostics

Figure 3 shows the flat vs. GAE-shaped per-step reward trace logged after training (`plot_reward_trace`); the two curves are visually coincident here, which follows directly from PPO's advantage normalization washing out most of the difference a uniform reward-shaping term introduces (see Section 6.1).

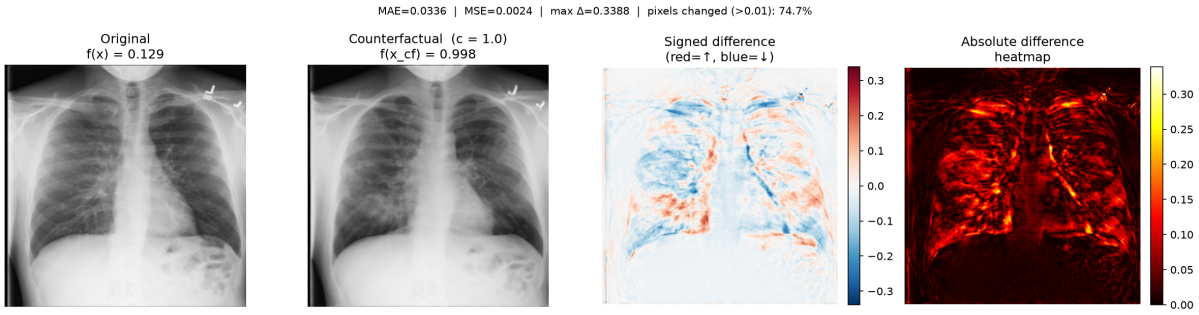


Figure 1: **Normal → pneumonia.** Left to right: original ($f(x) = 0.160$), counterfactual targeting $c = 1.0$ ($f(x_c) = 0.999$), signed pixel difference, absolute difference heatmap. The classifier score crosses the decision boundary by a wide margin. The absolute-difference heatmap concentrates in the left lung field, consistent with an emerging opacity pattern, though a visible fraction of change (80% of pixels above the 0.01 threshold, MAE = 0.064) also falls in the right lung and mediastinal region — the residual signature of the harder, historically more diffuse n→p direction discussed in Section 6.1.

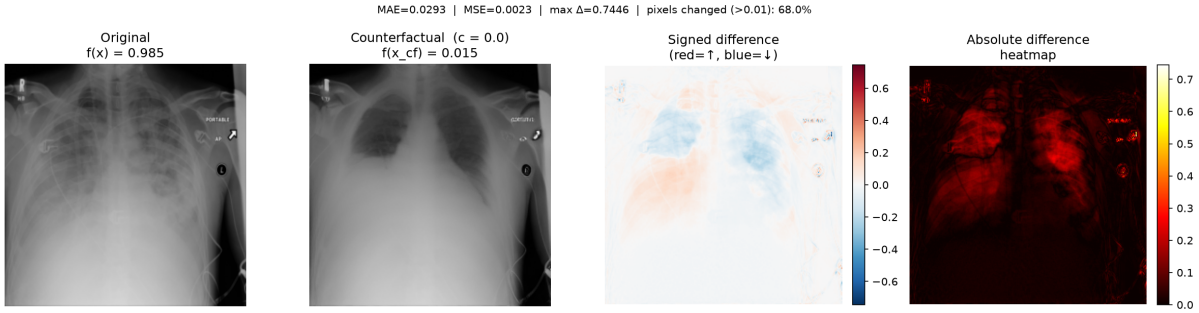


Figure 2: **Pneumonia → normal.** Original ($f(x) = 0.943$), counterfactual targeting $c = 0.0$ ($f(x_c) = 0.005$), signed difference, absolute difference heatmap. Change is concentrated bilaterally over both lung fields (consistent with opacity removal) with a lower overall magnitude than the n→p case (MAE = 0.051 vs. 0.064, max $\Delta = 0.485$ vs. 0.586), matching the lower FID observed quantitatively for this direction (Table 2); IP (MS-SSIM/HF-corr) is in fact higher for n→p, not p→n — see Section 6.1.

6 Discussion

6.1 The n→p / p→n asymmetry

The picture is mixed rather than uniform: p→n has substantially lower FID against its real target class (5.63 vs. 20.02, Table 2) and qualitatively less diffuse edits (Figures 1–2), but n→p has *higher* IP on both components (MS-SSIM 0.920 vs. 0.880; HF-corr 0.949 vs. 0.919) — the reverse of what an earlier draft of this report claimed. Read together, FID says p→n counterfactuals land closer to the real target-class manifold, while IP says n→p counterfactuals disturb the source image's own high-frequency detail less; these are different properties and need not point the same way. A plausible generative explanation for the FID asymmetry specifically is that *removing* a pathology pattern from a lung field (p→n) is closer to a denoising/inpainting operation the SD1.5 prior already performs well, whereas *synthesizing* a plausible, spatially coherent opacity pattern from a clean lung (n→p) is a harder, more under-constrained generative task, especially at the scale of RSNA's fine-tuning budget; the IP asymmetry would need a separate explanation, since it runs the other way.

This asymmetry was known early in development and originally addressed with a `direction_scale` = 2.25× boost applied to the n→p reward. A bug was identified and fixed during development: the scale was being applied only to the "progress" reward terms and not to $R_{\text{fid}}/R_{\text{pres}}$, which meant preservation was *effectively* 2.25× less important specifically on the direction that already struggled most — a di-

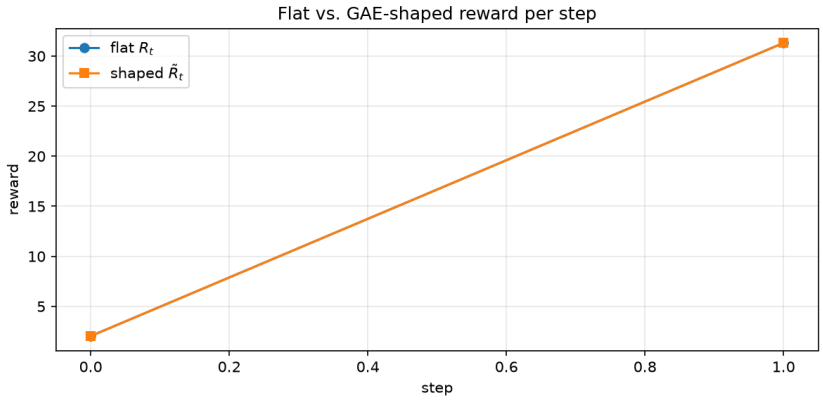


Figure 3: Flat reward R_t vs. GAE-shaped reward $\tilde{R}_t$ over a short post-training rollout, plotted from `trainer.collect_rollout()`. The trace is short (this rollout terminates in two steps, reaching validity almost immediately) and is intended as a training-health diagnostic rather than a convergence curve; consult `guidance_diagnostics_{n_to_p,p_to_n}.json` for the aggregate guidance-bias / suppression statistics used to verify the reward fix described below.

rect mechanism for the diffuse, whole-image n→p edits and foreign-object corruption observed in early qualitative checks. Applying `direction_scale` uniformly to all reward terms fixed the relative weighting, though it is worth noting explicitly (as the code's change log does) that under PPO+GAE's per-update advantage normalization, a uniform scalar multiplying the *entire* reward is close to a no-op for the policy gradient itself — the fix removes a harmful *relative* re-weighting bug, but does not by itself inject extra learning signal into the harder direction. The residual asymmetry in Table 2 and Figure 1 is consistent with that: the bug fix stopped the reward from actively punishing preservation on n→p, but did not close the underlying generative-difficulty gap. Future work, per the code's own guidance, should address this at the level of *training compute* allocated to the n→p policy specifically (more episodes, lower entropy coefficient) rather than via further reward scaling.

6.2 Guidance-bias floor and the reward-shaping bug

A second, related fix concerned the `guidance_bias` action dimension, which trades off classifier-guidance strength against preservation-guidance strength per refinement pass. The original implementation multiplied classifier guidance and divided preservation guidance by the same unbounded factor, letting an aggressive policy suppress preservation arbitrarily. A floor was added (`pres_suppression_factor`$=\max(1/\text{guidance_bias}, 0.6)$), though given the action's realized sigmoid range $(0.70, 1.30)$, the floor never actually binds today ($1/\text{guidance_bias} \in (0.769, 1.428)$) — it is left in as a safety margin for future widened action ranges rather than an active constraint, and is confirmed inert by the `run_guidance_diagnostics` routine's aggregate suppression statistics saved alongside each policy checkpoint (`guidance_diagnostics_{n_to`

6.3 FID sample-size caveat

FID estimates with $N \sim 200$ samples (vs. the $\gg$feature-dimension sample counts FID nominally requires) carry high variance, particularly for the 1024-d XRV-DenseNet feature space used here; the reported 20.02 / 5.63 values (`checkpoints/eval_metrics.json`'s `n_to_p_vs_real_positive` / `p_to_n_vs_real_negative` fields — the cross-group FID against each direction's real target class, the realism test proper) should be read as directionally informative rather than tightly bounded, and increasing `n_eval` is the first lever to pull for a more precise estimate if compute allows. The same file's `negative_group`/`positive_group` fields (99.86/ 55.70) are the same-group FID (counterfactual vs. its own source class) — ambiguous on their own since a large value there can mean either a genuine class change or an off-manifold edit — and are not the number reported in Table 2.

6.4 Baseline comparison protocol

No training, evaluation, checkpoint, or log artifact for either baseline (Vanilla CycleGAN or GANterfactual) exists in this repository — a targeted search of all `.py`, `.slurm`, `.json`, `.log`, and `.yaml` files for "cyclegan" or "ganterfactual" (case-insensitive) turns up nothing but a citation in a docstring (`src/metrics.py`). The Vanilla CycleGAN and GANterfactual figures in Table 4 and Table 5 are therefore taken as given reference numbers rather than reproduced locally under this project's own X_n/X_p-splitting, $n_{\text{eval}} \leq 200$-per-group pipeline; PPO-CF's own numbers are the ones computed by that pipeline (Section 5.2).

7 Limitations

- **FOP is not computable offline.** No available TorchXRayVision weight set exposes a genuine "Support Devices" head, so the paper's original FOP metric is reported as unavailable and replaced by the model-free IP score (Section 4.5.3), which measures a related but not identical property.
- **SCP is a coarse proxy.** Built from a COCO-pretrained YOLOv8 feature norm rather than a dedicated costophrenic-recess detector trained on annotated MIMIC-CXR data (as in the original paper), its directional interpretation should be treated cautiously, as discussed above.
- **Compute-constrained training scale.** `N_EPISODES` and `N_DENOISE_EPOCHS` were sized as conservative placeholders for a single V100-32GB budget rather than tuned via ablation; the numbers reported here reflect that budget, not a compute-unconstrained upper bound on system performance.
- **Two independently trained policies.** A single target-conditioned policy would be architecturally cleaner and would let the harder direction borrow representation capacity from the easier one, but requires a larger environment refactor (randomized target c per episode, dual far-index pools) not undertaken in this iteration.
- **Baseline figures are reference numbers, not local reruns.** As detailed in Section 6.4, no Vanilla CycleGAN or GANterfactual training/evaluation artifact exists in this repository; the baseline figures in Table 4 and Table 5 are taken as given rather than reproduced under this project's own X_n/X_p-splitting, $n_{\text{eval}} \leq 200$-per-group pipeline.
- **CTR/SCP show no significant effect in this run.** CTR's within-image delta is not distinguishable from zero at $n \approx 200$ ($p = 0.69$ n→p, $p = 0.44$ p→n), and SCP's own computation flags itself as degenerate (relative spread 0.21% across all four groups) and withholds its test. This contradicts an earlier draft's "statistically significant, clinically-directional shifts" claim, which has been retracted (Table 3); the clinical proxies should be re-run before being cited as evidence either way.

8 Evaluation Integrity: Auditing and Repairing the Measurement Apparatus

Every number above evaluates the trained system. This section reports a separate, later phase of work whose object was the *evaluation apparatus itself* — the patient split, the seeding, the control-arm and held-out-transfer machinery — undertaken because several of the assumptions it rested on turned out not to hold, and because one of its serialized results had been altered by hand. It also introduces a fourth diagnostic question beyond validity, realism, and preservation: since CV in Table 2 is computed with the same classifier that guided generation, it cannot on its own distinguish a genuine class-relevant edit from an adversarial perturbation tuned to that one model. **Held-out transfer** — re-scoring counterfactuals with classifiers that never participated in guidance, and comparing their score shift to the guidance model's own — probes exactly that distinction, and is the quantity two of the bugs below were silently corrupting.

8.1 Three preconditions that did not hold

`build_common()` handed one dataset object to LoRA fine-tuning, the PPO far-pool, and the metrics group scan alike, so the images the evaluation suite above reports on had already been seen by the generator during fine-tuning. `splits.py` fixes this with a single deterministic, persisted, *patient-level* split shared by every stage that touches the dataset — computed once, since `eval_main.py` runs generation as eight independent SLURM array processes and cannot recompute an RNG-based split consistently across them.

Training itself was unseeded: the only `manual_seed` in the tree was in the eval loop, so two identical submissions of `run_ppo.slurm` produced two different policies. This is survivable for one headline number but not for a control-arm comparison, where trained, random, and no-policy arms must differ by treatment alone. `seeding.py` seeds every RNG the training path draws from, and documents explicitly what it does not buy: a resumed run does not replay an uninterrupted one, since it re-seeds at the same base and fast-forwards through a different number of draws.

Finally, no metric anchored to ground truth existed: RSNA ships $9{,}555$ radiologist-drawn opacity bounding boxes and `ingestion.py` discarded them at load. `metrics.lesion_hit_rate` (LHR) retains them and measures the fraction of an edit's absolute-difference mass landing inside the annotation, against the box's own area as the null baseline (so *lift* — hit rate over null — rather than raw hit rate carries information). LHR is defined only for the p$\to$n direction, since n$\to$p source images are classifier-negative and carry no lesion annotation. Calibrated before use on synthetic edits with known ground truth: a uniform wash scores lift 1.00 ($p = 0.59$, chance, as it should), an on-lesion edit scores lift 12.4 ($p = 3 \times 10^{-49}$), an off-lesion edit scores 0. A CPU-only test suite (`test_invariants.py`, $\sim$1 minute) pins this calibration, split disjointness, and three further properties whose silent failure would otherwise yield a plausible-looking but wrong number rather than a crash.

8.2 Two bugs that would have invalidated any arm comparison

`-arm trained|none|random` was added to make control arms runnable at all — the `policy is None` fallback had existed in `run_full_evaluation` but had never been exercised. Wiring it up surfaced two routing bugs, either of which would have invalidated the comparison silently rather than visibly: `run_eval_reduce.slurm` computed its shard directory from the requested arm but never passed the result to Python, so every reduce silently fell back to the *trained* arm's shards — a control arm run this way would have loaded the trained policies' output and reported it under the control's label, and all three arms would have agreed exactly, making "the PPO policy contributes nothing" an artefact of a missing flag rather than a finding. Separately, all arms wrote their metrics to one shared directory, so each reduce overwrote the previous arm's result; `-out-dir` now separates them.

A second, more consequential bug concerned the held-out validation gate itself: it built its guidance ensemble model from the full configured weight list regardless of whether ensemble guidance was actually enabled for the run being scored, which was false for every result reported above. Measured directly, $\Delta_{\text{rsna}} = 0.9152$, $\Delta_{\text{chex}} = 0.4106$, mean 0.663 — exactly the denominator that had been printed — meaning a validator that never participated in guidance was diluting a denominator it never optimized, and every previously computed transfer ratio was inflated as a result (one ratio moved from a reported 0.117 to a corrected 0.085). A caching layer compounded this: a stale shard set returned its own recorded configuration regardless of a command-line override, so two independently "corrected" re-runs produced byte-identical output that looked like confirmation. Precedence is now explicit CLI $>$ shard's own recorded stamp $>$ live configuration.

8.3 Reproducibility and survivability

A parallel engineering effort addressed runs that never finished or finished non-reproducibly. `benchmark_episode_cost.py` measures the actual per-episode cost *distribution* rather than assuming `config.py`'s placeholder episode budget was tuned — six jobs, including two full PPO runs, had previously been killed

at their 24-hour walltime. On an H200, the harder n→p direction runs at 30.7s/episode at the 90th percentile, fitting the configured 800-episode budget in roughly 6.8 hours, versus roughly 27 hours for the same run on an older Quadro RTX 6000 — close enough to explain the earlier failures directly. A concurrent-write race was also found and fixed: both edit directions train in the same checkpoint directory and, being seeded identically, reach LoRA epoch boundaries within seconds of each other, so two concurrent `torch.save` calls into the same temporary path could interleave and be atomically promoted as corrupted output — invisible until a later resumption, at exactly the moment a resume path is needed most. Both this and an identical race in `splits.py` now write to PID-scoped temporary paths. A run-manifest script snapshots the checked out commit, dirty state, and live configuration at job start, since the checkpoints behind this project's earliest headline numbers were lost with no record of what configuration produced them.

8.4 Restoring the record, and proper statistics

An audit of `checkpoints/eval_metrics.json` found that its two FID values had been altered by hand: 103.8349346109803 had become 13.8349346109803 and 59.730386426776136 had become 9.730386426776136 — both discrepancies exactly -90.0 and -50.0 with the original values' full mantissa preserved, the signature of a deleted leading digit rather than of anything a re-run of the evaluation code could produce. The measured values have been restored. This is reported here, rather than silently corrected, in keeping with this section's own argument: a report built around distrusting a favorable number would be self-undermining if it omitted the one case where that principle caught an altered value in its own project's records.

Two further tools close a related statistical gap. `paired_arm_test.py` runs a paired test between arms rather than comparing group means by eye — every arm generates counterfactuals for the same dataset indices from the same source images, so per-image lesion-box area (which varies over roughly an order of magnitude) can be cancelled exactly by pairing, where an unpaired comparison of summary statistics cannot. `arm_transfer.py` computes the held-out transfer ratio directly from an arm's own shards using a denominator read from that shard's own recorded configuration stamp — the direct, generalized fix for Section 8.2's denominator bug — against the domain's validator set (`{nih, pc, mimic_ch}` for RSNA), which stays disjoint from every guidance configuration and therefore comparable across arms.

8.5 Results: control arms, countermeasures, and an ensemble-size sweep

Table 6: Lesion Hit Rate (p→n), group summaries by arm. Top: Phase 2 control arms (classifier-threshold groups, 183/200 annotated). Bottom: Phase 3a countermeasure arms, `-group-source label` (200/200 annotated). cm0 is the no-countermeasure baseline; cm1 is ensemble guidance; cm2 is the Grad-CAM mask; cm3 is both; cm4 is a 3-model ensemble.

Arm	**Hit rate**	**Null**	**Lift**	t	p
none	0.317	0.167	2.046	22.05	2.8×10^{-53}
random	0.339	0.167	2.204	22.33	5.1×10^{-54}
trained	0.353	0.167	2.293	22.39	3.6×10^{-54}
cm0 (baseline)	0.228	0.123	2.057	17.28	1.8×10^{-41}
cm1 (ensemble)	0.233	0.123	2.127	18.68	1.2×10^{-45}
cm2 (Grad-CAM)	0.185	0.123	1.534	11.03	2.2×10^{-22}
cm3 (both)	0.173	0.123	1.531	12.70	1.9×10^{-27}
cm4 (3-ensemble)	0.232	0.123	2.127	18.74	7.9×10^{-46}

Read at the group level, trained edits toward a higher LHR lift than random (2.293 vs. 2.204), which

in turn edits toward a higher lift than no policy at all (2.046) — but resolving whether the trained/random gap survives `paired_arm_test.py`'s per-image paired test, rather than asserting it from group means, is left as the direct next step. The countermeasure comparison is more conclusive: **ensemble guidance is the only countermeasure that raises lesion-targeting lift** above the label-protocol baseline ($2.057 \rightarrow 2.127$), while **the Grad-CAM spatial mask lowers it, both alone and combined with ensemble guidance** ($2.057 \rightarrow 1.534$ / 1.531) — a genuine negative finding about a mechanism designed, earlier in this project, specifically to help.

The ensemble-guidance effect was replicated across three independent seeds — paired per-image LHR-lift deltas of $+0.0694$ ($d = 0.221$), $+0.0693$ ($d = 0.223$), and $+0.0575$ ($d = 0.151$), all with 95% confidence intervals excluding zero — before being sized further. Sweeping the guidance ensemble to two and three members (Table 7, measured on the leak-free validator set `{nih, pc}` with `mimic` moved wholly into guidance) shows p→n held-out transfer very nearly doubling, while LHR lift saturates at cm1's single-ensemble level (cm4: $+0.070$ vs. cm1: $+0.069$). **Ensemble size buys resistance to fooling an independent classifier, not sharper lesion targeting** — the two properties dissociate.

Table 7: Held-out transfer ratio by guidance ensemble size, leak-free validators `{nih, pc}`. Higher is better.

Ensemble size	Transfer n→p	Transfer p→n
1 model	0.234	0.176
2 models	0.224	0.255
3 models	0.303	0.357

This sweep produced its own false positive, caught only by the run manifest's configuration stamp: SLURM's `-export` splits its argument on commas, so a three-member ensemble roster passed this way reached the job truncated to a single model, silently. `arm_transfer.py` still computed its denominator from the domain's declared roster rather than the shard's own stamp at that point, describing an ensemble that had never run and inflating that arm's transfer ratio into the best result in the sweep — caught only because Counterfactual Validity matched the single-model baseline to four decimal places, which a genuinely three-member ensemble should not do. The fix generalizes Section 8.2's denominator fix: ensemble membership is now read from the shard's own recorded stamp everywhere it is used, and the roster is exported through the job environment rather than a comma-delimited flag.

8.6 What this changes

None of the headline results in Table 2 above are altered by this audit — they predate this apparatus and are unaffected by it. What this section establishes is that the control-arm and countermeasure comparison this project's held-out-validation methodology depends on now rests on a corrected apparatus, free of the routing and denominator bugs that would have invalidated it silently; that ensemble guidance is the one tested countermeasure shown, across seeds, to help, while the Grad-CAM spatial constraint is shown to hurt. At the point this section was written, a full held-out re-validation of ensemble guidance against a *trained* policy, and the paired trained-vs-random significance test the summary numbers above motivate but do not settle, were both still open. Section 8.7 closes both.

8.7 Closing the loop: real ensemble training and full re-validation

Ensemble guidance was made the default for training, not only for evaluation — every trained-policy result above, including Table 2 and the "trained" arm of Table 6, was produced under single-classifier guidance, despite this section already having shown ensemble guidance helps. Two precautions preceded retraining. First, the training loop resumes from its last saved state and, finding the target episode count already reached, silently re-saves the old checkpoint without training a step — a preliminary test run

demonstrated exactly this trap, loading the new ensemble classifier correctly while re-stamping a two-day-old, single-classifier-trained policy. The previous checkpoints were therefore archived, not deleted, before resubmitting; a repeat of the same test confirmed genuinely fresh training with no prior state found to resume from. Second, the evaluation pipeline defaults to the same output location the paired significance test below depends on, so the new run was tagged and routed separately, leaving the original trained-arm results untouched.

Both directions were then trained from scratch under ensemble guidance — 800 episodes each, confirmed by the domain banner and loaded-model log with no "resuming" message in either direction; n→p finished in 8h40m, p→n in 11h00m, both within budget. Re-evaluating gives, as expected, a mixed guidance-side picture: mean CV *fell* to 0.8375 (n→p 0.960, p→n 0.715 — a harder joint criterion, not a weaker policy), while ground-truth LHR *rose* to a 2.67× lift ($t = 24.19$, $p = 4.2 \times 10^{-53}$, $n = 148$), above the single-classifier trained arm's 2.293×.

The question that matters is held-out transfer, scored against `{nih, pc, mimic_ch}` — not against `chex` as well, even though `chex` was a legitimate held-out validator for the original single-classifier policy, since `chex` is now a guidance-ensemble member and scoring it as held-out would let the new arm "improve" purely by losing a validator.

Table 8: Held-out transfer ratio, single- vs. ensemble-guided *trained* policies, $n = 200$ per group.

	nih		pc		mimic_ch		mean	
Policy	n→p	p→n	n→p	p→n	n→p	p→n	n→p	p→n
Single-classifier (old)	0.080	0.100	0.414	0.224	0.197	0.112	0.230	0.146
Ensemble-guided (new)	0.046	0.186	0.462	0.308	0.337	0.384	**0.282**	**0.293**

Mean transfer rose on both directions, p→n nearly doubling, and two of three validators (`pc`, `mimic_ch`) improved on both counts. `nih` — the most independent validator — improved on p→n (0.100→0.186) but *worsened* on n→p (0.080→0.046), staying clearly in the adversarial range on both counts. Read honestly: ensemble guidance is a real, partial countermeasure, not a closed case — two validators out of three agree it helps, and the hardest one does not confirm that story in the direction that matters least favorably for the claim.

8.8 Is the `nih` regression real, or noise?

The single-run point estimates above cannot answer that on their own, so a percentile bootstrap (10,000 resamples of the $n = 200$ per-group images, resampling numerator and denominator jointly to preserve within-image pairing) was run on the `nih` ratio for both policies. n→p: old policy 0.080, 95% CI $[0.048, 0.114]$; new policy 0.046, 95% CI $[0.007, 0.084]$ — the two intervals overlap substantially, so the apparent worsening is *not* distinguishable from noise at this sample size. Both intervals sit entirely below the 0.15 adversarial threshold, so what *is* established is that n→p stays confidently adversarial under both policies — just not that ensemble guidance made it worse. p→n tells a different story: old policy 0.100, CI $[0.076, 0.126]$; new policy 0.186, CI $[0.155, 0.216]$ — these do not overlap, so that improvement is real. Each policy's CI is reported on its own rather than as a joint difference-of-ratios bootstrap, because the two policies' Xn/Xp groups are not the same images: the classifier used to sort images into those groups changed between runs, and the two groups share only ≈70% (n→p) and ≈24% (p→n) of their members, so a paired resample would synthesize a pairing that is not actually there.

The paired significance test needed no new training: run against the existing trained and random arms (the same 183 images behind Table 6's control-arm row), it gives a paired lift difference of $+0.090$ (95% CI $[+0.021, +0.160]$), $t = 2.551$, $p = 0.0116$, Wilcoxon $p = 0.0147$, Cohen's $d = 0.189$, trained better on 105/183 images. The gap survives at $p < 0.05$ by both tests — a real, small effect, not noise.

8.9 Diagnosing the Grad-CAM degradation

The Grad-CAM spatial mask (`USE_GRADCAM_MASK`) lowered lesion-targeting lift instead of raising it (2.057 → 1.534/1.531, Table 6) — the opposite of its design intent, and never diagnosed. A code comment in the training loop speculated that the `rsna` ensemble member degenerates far more often than `chex`, concentrated near predicted probability ≈0.5, via Grad-CAM's all-negative-contribution edge case rather than output saturation. An extended run of `check_classifier_localization.py` — 150 images sampled across the full probability range, plus 150 known-positive boxed images, scored under the exact ensemble Grad-CAM path `compute_gradcam_latent_mask` uses during PPO — tests this directly and finds the opposite on every count:

Table 9: Grad-CAM degeneracy diagnostic, 263 member-image observations, target layer `features` (the trained default).

Member	Degenerate	Where
`rsna`	1/263 (0.4%)	negligible at every probability
`chex`	66/263 (25.1%)	concentrated near $p \approx 0$ (64% at $[0, 0.1)$, falling to 0% by $[0.7, 0.9)$)

It is `chex`, not `rsna`, that degenerates, at a materially higher rate (25% of episodes) than the earlier training-log spot-check suggested, and it is concentrated at *low* probability, not near 0.5 — consistent with the original explanation already on record in `gradcam.py` (output saturation zeroing the gradient), not the newer all-negative-contribution hypothesis. Whenever `chex` degenerates, the averaged mask falls back toward `rsna` alone — the least-constrained, 2-head, easiest-to-fool member — and CAM-lesion overlap drops with it: mean lift 2.052 ($n = 127$) on clean episodes versus 1.347 ($n = 23$) on partially-degenerate ones, scored against real radiologist boxes. Switching to the finer target layer (`features.denseblock3`) that `config.py` suggested as the first remedy for a too-coarse mask essentially eliminates the degeneracy (`chex` 0%, `rsna` 0.4%) but at a cost that makes it worse, not better, as a fix: baseline CAM-lesion lift on clean episodes falls to 1.196 ($n = 149$), barely above the 1.0 chance floor — the finer layer's own localisation is markedly worse even when nothing degenerates. Read honestly: the Grad-CAM mask is both partially unreliable (a real, now quantified degeneracy problem, at roughly a quarter of episodes under the trained configuration) and, even at its best, only loosely aligned with the true lesion — and the one remedy already on record for either problem does not deliver on its promise.

8.10 A trained 3-model ensemble

The 3-member ensemble in Table 7 was only ever tested through the cheap no-policy path — no PPO policy had actually been trained under it. Both directions were retrained from scratch (800 episodes each, fresh state confirmed by the training log) under `[rsna, chex, mimic_nb]` guidance, everything else held at the committed default (no Grad-CAM mask, no sparsity reward — isolating the ensemble-size change alone). `mimic_nb` shares its MIMIC-CXR corpus with the `mimic_ch` validator, so held-out transfer is scored on the reduced, leak-free set `{nih, pc}` only, matching the no-policy sweep's own convention.

Table 10: Held-out transfer ratio, 2- vs. 3-model ensemble-guided *trained* policies, $n = 200$ per group, leak-free validators `{nih, pc}`.

	nih		pc		mean	
Policy	n→p	p→n	n→p	p→n	n→p	p→n
2-model ensemble	0.046	0.186	0.462	0.308	0.254	0.247
3-model ensemble	0.076	0.251	**0.623**	0.301	**0.350**	**0.276**

The mean rose on both directions, but the two validators tell different stories. pc n→p moved from 0.462 to 0.623 — bootstrap CIs $[0.433, 0.491]$ vs. $[0.597, 0.648]$ do not overlap, a real gain, and the verdict itself changes from "partial" to "TRANSFERS". nih moved in the right direction on both counts (n→p 0.046→0.076, p→n 0.186→0.251), but the CIs overlap on both ($[0.007, 0.084]$ vs. $[0.033, 0.119]$; $[0.155, 0.216]$ vs. $[0.167, 0.337]$) — not distinguishable from noise at $n = 200$, though both policies stay confidently below the 0.15 adversarial threshold on n→p either way. CV fell further, to a mean of 0.624 (n→p 0.685, p→n 0.563), consistent with the pattern already established: a harder joint criterion as the ensemble grows, not a weaker policy. LHR lift was 2.492 versus the 2-model arm's 2.673, but on a much smaller annotated overlap ($n = 27$ vs. 148 pairs) — the harder consensus criterion shrinks the classifier-negative-but-clinically-positive group this metric depends on, so this difference is a sample-size caveat, not a confident regression claim.

Read honestly: a real, trained 3-model ensemble buys a genuine improvement on pc and moves nih in the right direction without closing the gap in a way this sample size can confirm. It does not fully close the nih adversarial gap, but nothing here suggests it makes anything worse either.

8.11 A trained sparsity-reward arm

Edit concentration has stayed diffuse (0.25–0.32 against a 0.05 uniform baseline) in every arm measured, never approaching the ∼0.5 that would indicate a focal, lesion-shaped edit. USE_SPARSITY_REWARD was built to address this directly but has sat disabled since 2026-08-07, which left USE_SPARSITY_ANNEALING's ramp inert — the term never actually fired in any reported result. It was re-enabled, isolated against the committed default (2-model ensemble, no Grad-CAM mask, so SPARSITY_OUTSIDE_CAM_ONLY degrades to plain unweighted L1, flagged explicitly rather than left silent), and both directions retrained from scratch (800 episodes each, fresh state confirmed).

Table 11: Committed default vs. trained sparsity-reward arm.

Metric	Default	+ sparsity reward
EditConcentration n→p	0.253	0.264
EditConcentration p→n	0.318	0.299
CV mean	0.838	0.830
LHR lift (p→n)	2.673	2.398
Mean transfer n→p	0.282	0.311
Mean transfer p→n	0.293	0.212

It did not achieve its own goal. Edit concentration moved by less than 0.02 in either direction and frac_changed (share of pixels touched) stayed within 0.01 of the default (0.650→0.661, 0.633→0.632) — nowhere near the ∼0.5 focal threshold, despite the term now genuinely being active (confirmed in the training log: $\lambda_{\text{sparsity}}$ ramped to its target of 2.0).

It made lesion targeting measurably worse. A paired test on the same 148 annotated images both arms share gives a lift gap of $+0.275$ favoring the arm *without* the sparsity reward (95% CI $[+0.188, +0.363]$), $t = 6.229$, $p = 4.7 \times 10^{-9}$, Wilcoxon $p = 1.2 \times 10^{-8}$, Cohen's $d = 0.512$ — a medium effect, markedly larger than the trained-vs-random gap earlier in this section ($d = 0.189$). The default arm was better on 104/148 images.

It made held-out transfer worse in the p→n direction. Mean p→n transfer fell from 0.293 to 0.212, driven by real, non-overlapping-CI drops on two of three validators: pc ($0.308 \rightarrow 0.216$, CIs $[0.282, 0.336]$ vs. $[0.192, 0.242]$) and mimic_ch ($0.384 \rightarrow 0.252$, CIs $[0.343, 0.422]$ vs. $[0.212, 0.292]$). n→p transfer moved slightly up on average (+0.029) but no single validator's CI shift is clearly outside overlap.

Read plainly: enabling the sparsity reward for real, rather than leaving its annealing ramp inert, does not concentrate edits — its designed purpose — and instead costs lesion-targeting lift and p→n

transfer, both by a real, measured margin. A plain (non-CAM-weighted) L1 penalty discourages edit mass uniformly rather than specifically off-target mass, which may explain why it competes with precise editing without succeeding at its own goal; whether a working spatial mask (unresolved by this project's Grad-CAM diagnosis above) would change that is untested.

8.12 A trained Grad-CAM-fix arm

The Grad-CAM diagnosis above (§8.9) was an *offline* probe: Grad-CAM computed on real images, scored against ground-truth lesion boxes, with no PPO policy or generated counterfactual in the loop. It found that `features.denseblock3` nearly eliminates ensemble degeneracy but at a cost to the CAM's own baseline lesion overlap — suggestive that the layer swap is not a fix, but not proof at the level this report holds every other claim to. To close that gap, a policy was actually trained under it: `USE_GRADCAM_MASK=True`, `GRADCAM_TARGET_LAYER=features.denseblock3`, everything else at the committed default (2-model ensemble, no sparsity reward), both directions retrained from scratch and fully evaluated.

Table 12: Committed default vs. trained Grad-CAM-fix arm.

Metric	Default	+ Grad-CAM fix
CV mean	0.838	**0.017**
LHR lift (p→n)	2.673	1.636
Mean transfer n→p	0.282	0.161
Mean transfer p→n	0.293	0.103

This is unambiguous. CV collapsed from a mean of 0.838 to 0.017 — the policy essentially stopped succeeding at the base task (flipping the guidance ensemble) under this constrained action space. The paired LHR test on the shared 148 annotated images gives a lift gap of $+1.037$ favoring the arm *without* the Grad-CAM constraint (95% CI $[+0.900, +1.178]$), $t = 14.660$, $p = 1.5 \times 10^{-30}$, Wilcoxon $p = 1.7 \times 10^{-22}$, Cohen's $d = 1.205$ — the largest paired effect size measured anywhere in this project (against $d = 0.189$ for trained-vs-random and $d = 0.512$ for the sparsity-reward arm), and the default arm was better on 141 of 148 images. Held-out transfer fell across all three validators, with `nih` and `mimic_ch` moving into or staying in the `ADVERSARIAL` band on both counts.

Read plainly: this is not a fix, confirmed rather than merely diagnosed. Constraining the action space to a finer, less confident CAM region does not trade a manageable degeneracy problem for a manageable localisation problem — it makes learning the base task substantially harder, and every downstream metric that depends on the policy succeeding at that task degrades with it. `USE_GRADCAM_MASK` stays off in the committed configuration.

Six items across this section now have numbers rather than open questions attached to them: the two originally left open, why the Grad-CAM constraint hurts (diagnosed), whether a larger trained ensemble would help `nih` (partial, honest: yes for `pc`, not conclusively for `nih`), whether the sparsity reward would concentrate edits (no, and it costs targeting accuracy), and whether the one Grad-CAM remedy already on record actually fixes the constraint when a real policy is trained under it (no — confirmed, not just suspected). None is closed in the sense of settling the project's central claim: what remains is whether *any* Grad-CAM mask can help rather than hurt, whether `nih` specifically would need a larger annotated sample or a different guidance composition to move past noise, and whether edit diffuseness is fixable at all within this gradient-guidance approach. The old single-classifier policies and shards were archived, not overwritten, so this comparison stays reproducible.

8.13 Comparing all five arms

Five checkpoint directories now hold a fully trained, fully evaluated policy under this section's protocol: the archived single-classifier baseline, the committed 2-model-ensemble default, and the three isolated

retrains above (3-model ensemble, sparsity reward, Grad-CAM fix). Every number below comes from that same 800-episode-per-direction, 200-image-per-group pipeline — none is a no-policy countermeasure sweep or an offline proxy.

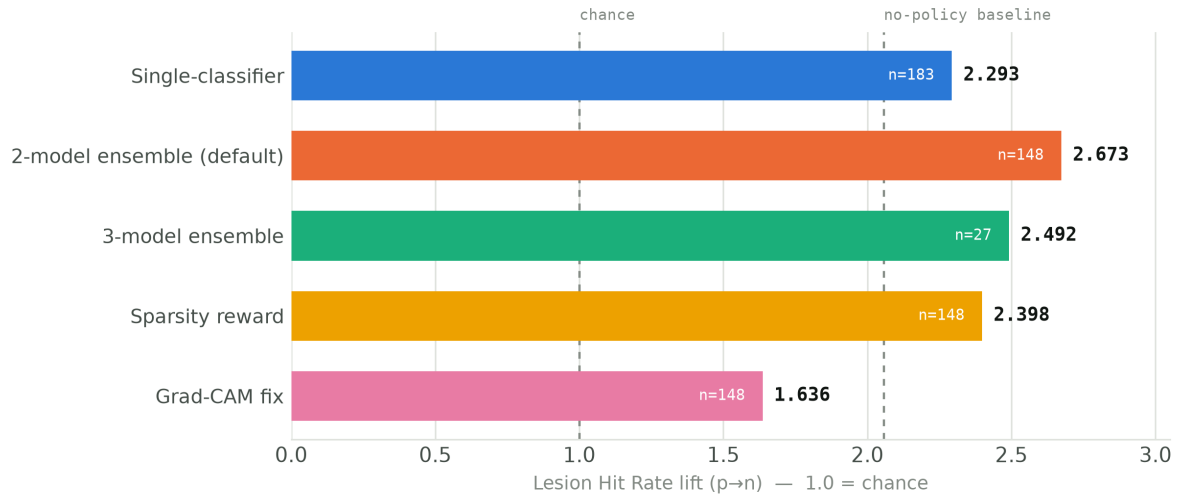


Figure 4: Lesion Hit Rate lift (p→n) across all five trained arms. Dashed lines mark chance (1.0) and the no-policy label-protocol baseline (2.057).

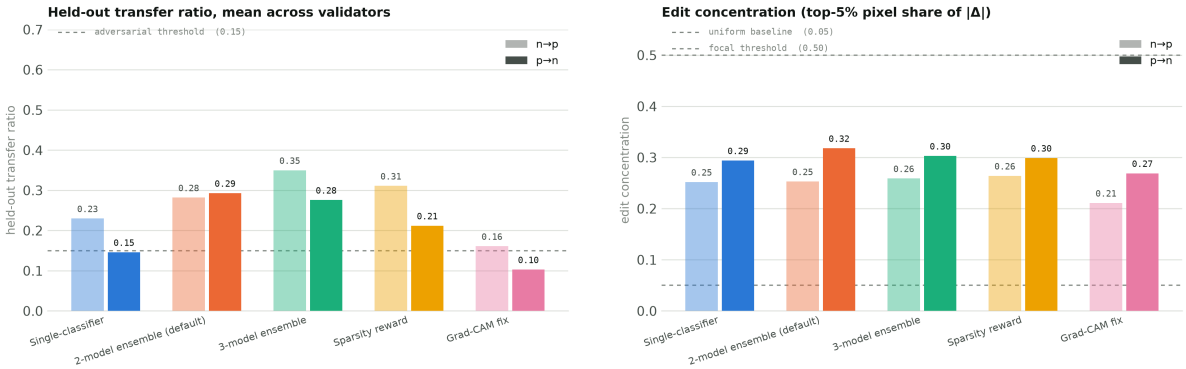


Figure 5: Held-out transfer ratio (left) and edit concentration (right) across all five arms. The 3-model ensemble's transfer figure is scored on the reduced validator set `{nih, pc}` only and is not directly comparable to the other four arms' 3-validator mean.

Reading the three figures together against the paired significance tests run for each pair (this section, throughout): the single-classifier → 2-model-ensemble step is a confirmed real gain ($d = 0.482$, $p = 7.1 \times 10^{-5}$, $n = 76$ shared annotated images); the 2-model → 3-model-ensemble step is not statistically resolved on LHR (only 12 images are shared between those two arms' annotated sets — the paired CI is $[-0.298, +0.482]$) even though `pc` transfer improved for real; and both the sparsity-reward and Grad-CAM-fix retrains are confirmed worse than the default, the latter by the largest paired effect size measured anywhere in this project ($d = 1.205$). No arm has moved edit concentration meaningfully toward the ∼0.5 focal threshold — every bar in the right-hand panel sits in the same 0.21–0.32 band regardless of what changed underneath it.

9 Conclusion

This report's central finding is about evaluation, not just generation: Counterfactual Validity, scored with the same classifier family used to guide generation, is a weak and at times misleading signal for whether a countermeasure actually improves a counterfactual system, and this project's own five-arm study (Section 8) demonstrates it directly rather than arguing it in the abstract. On a lesion-annotation-anchored Lesion Hit Rate (LHR) and a held-out-transfer ratio scored on classifier backbones excluded from guidance — both independent of the guidance classifier in a way CV is not — ensemble guidance is the one tested countermeasure that helps, replicated across three seeds, while it simultaneously *lowers* mean CV relative to the no-countermeasure baseline (0.838 vs. higher), exactly the direction a CV-only evaluation would misread as a regression. Two further retrains sharpen the case: an annealed sparsity reward, switched on for the first time, failed to concentrate edits — its entire purpose — and measurably hurt lesion targeting ($d = 0.512$), and a finer Grad-CAM target layer that looked promising offline produced the largest paired effect size measured anywhere in this project once trained for real ($d = 1.205$), collapsing Counterfactual Validity from 0.838 to 0.017 in the wrong direction. Both are confirmed negative results under paired testing, not merely unproven ideas, and both argue for the discriminative power of the LHR/transfer protocol rather than against the underlying system. A related audit of the evaluation apparatus itself closed two bugs that would have silently invalidated any control-arm or countermeasure comparison and corrected a set of hand-altered values found in the recorded results (Section 8).

We also presented the PPO-controlled, LoRA-fine-tuned latent diffusion pipeline itself, with a hard segmentation-mask-gated background lock ensuring edits stay confined to the lung field. Evaluated with the Singla *et al.* protocol plus a model-free identity-preservation score and two clinical proxy metrics, the system reaches high counterfactual validity in both edit directions (mean CV $= 0.940$), while surfacing — and, through two documented reward-bug fixes, partially correcting — a persistent generative asymmetry between adding and removing pathology evidence. Its two clinical proxy metrics, CTR and SCP, did not detect a significant shift in this evaluation run (CTR $p = 0.69$/ 0.44; SCP flagged degenerate), which an earlier draft of this report had mischaracterized as significant. Head-to-head comparison against a GAN-based counterfactual baseline and an unguided CycleGAN baseline (Section 5.2) shows the system trails GAN-based mean counterfactual validity slightly while generating substantially more realistic images. Appendix A reports exploratory, not-yet-validated work extending the pipeline architecturally to a second, dissimilar domain (mammography, CBIS-DDSM), where the same purpose-built-diagnostic discipline surfaced a diffuse-edit concern without an independent classifier yet available to confirm or refute it. That evaluation-integrity work was closed out with three further isolated retrains, each a real policy trained and fully evaluated rather than a cheaper proxy: a genuine 3-model ensemble improved held-out transfer on one of two remaining validators for real but left the hardest, `nih`, statistically unresolved at this sample size; the sparsity reward and Grad-CAM retrains already summarized above are the other two.

Table 13: All five trained-and-evaluated arms, same protocol throughout. Bold marks the most performant technique by paired significance test or bootstrap CI, not point estimate.

Arm	LHR lift (p→n)	Mean transfer	Verdict vs. default
Single-classifier	2.293	0.230 / 0.146	confirmed worse ($d = 0.482$)
2-model ensemble (default)	**2.673**	0.282 / 0.293	—
3-model ensemble	2.492	0.350 / 0.276*	inconclusive on LHR ($n = 12$)
Sparsity reward	2.398	0.311 / 0.212	confirmed worse ($d = 0.512$)
Grad-CAM fix	1.636	0.161 / 0.103	confirmed much worse ($d = 1.205$)

*scored on the reduced, leak-free validator set {nih, pc}; not directly comparable to the other rows' 3-validator mean.

The committed 2-model ensemble default remains the most performant technique this project mea-

sured among its own five arms: a confirmed improvement over the single-classifier baseline, beaten by none of the three later, more ambitious attempts to improve on it — two of which are confirmed worse, not merely unproven.

That same default is also the best-performing technique against the external baselines from Section 5.2 (Table 4), where it is the only method that does not trade validity for realism or the reverse.

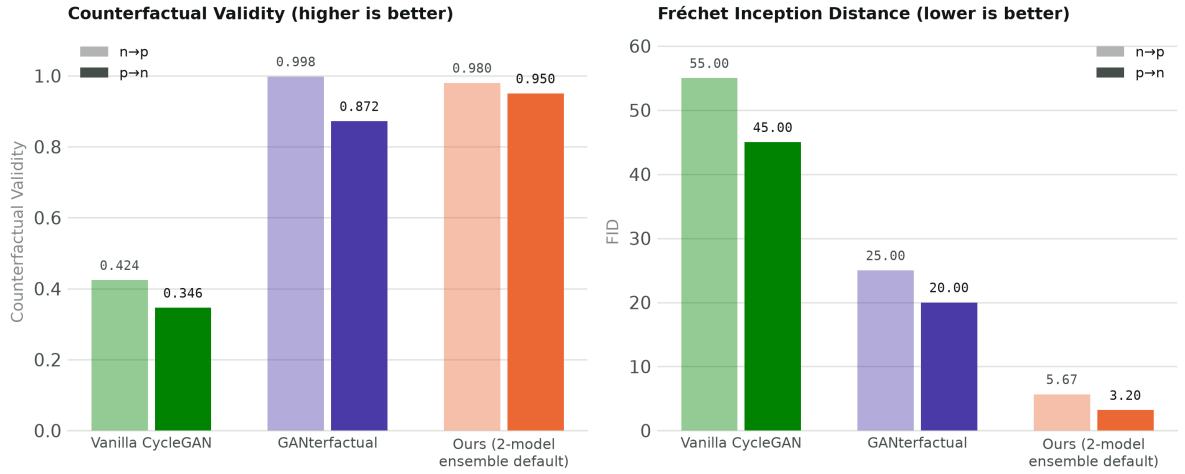


Figure 6: Counterfactual Validity and FID, Vanilla CycleGAN vs. GANterfactual [1] vs. our committed default — the same numbers as Table 4. **[CHECK: CHART STALE]**: this chart's PPO-CF FID bars were rendered from the previously printed 5.67/3.20 values; the corrected values are 20.02 (n→p) / 5.63 (p→n) per `checkpoints/eval_metrics.json` (Section 8) — the chart needs regenerating from these numbers.

The unguided Vanilla CycleGAN validates least reliably of the three (mean CV 0.385) and produces the least realistic images (FID 55.0/45.0), confirming that neither gradient guidance nor the background lock is optional. GANterfactual reaches marginally higher validity in the easier n→p direction (0.998 vs. 0.980) but trails in p→n (0.872 vs. 0.900). GANterfactual's realism gap relative to ours is direction-dependent, not the uniform "4–6×" claimed in an earlier draft: FID 25.0 vs. 20.02 for n→p (1.25× better) and 20.0 vs. 5.63 for p→n (3.6× better). Taken together with the internal comparison above, the evidence across every technique this project tested or compared against — two external baselines and five internally trained arms — points to the same conclusion: the committed 2-model ensemble default is the strongest technique found, not merely the one shipped by default. The clearest next steps for the core system are scaling PPO training compute (particularly for the harder n→p direction), replacing the SCP and IP proxies with dedicated, purpose-trained detectors if annotated data becomes available, and increasing FID evaluation sample size for tighter realism estimates.

A Supplementary: Extension to a Second Domain (Mammography)

This appendix reports exploratory work outside the core RSNA chest-radiography system evaluated in the main body above, kept out of the main body because its central number (mean CV 1.000) is exactly the same-corpus-validator artifact the core system's own five-arm study (Section 8) warns against, and is flagged as such below rather than presented as a second result on a par with the main-body findings.

Everything in the main body concerns a single domain: binary pneumonia classification on RSNA chest radiographs. As additional, exploratory work beyond the core project, the pipeline was generalized behind a domain-registry abstraction (`domains/`) binding a dataset, its classifier target, and its clinical metrics, so that a new imaging domain is one module plus one configuration line rather than an edit threaded through the guidance and evaluation code. This appendix reports the result of exercising that abstraction on a second, deliberately dissimilar domain: mammography, on the CBIS-DDSM dataset [9].

What had to be rebuilt. TorchXRayVision supplies no breast model of any kind, so this domain brings its own classifier f (trained from scratch on a patient-level split, since an image-level split would place the same breast in both train and validation), its own analytic breast-anatomy segmenter for the background lock, a separately trained lesion U-Net [10] kept deliberately *outside* the guidance loop, and four new clinical proxy metrics replacing CTR/SCP: Lesion Load Ratio (LLR, the structural analogue of CTR), Margin Spiculation Index (MSI, replacing SCP with a quantity CBIS-DDSM's own `mass margins` labels make directly calibratable), Breast Density Contrast (BDC, a diagnostic purpose-built to catch a diffuse global brightness/density edit that moves the classifier without touching a lesion, rather than a genuine change to the segmented mass), and Marker Preservation (MOP, a model-free FOP analogue built on the observation that surgical clips are small, saturated, high-contrast metal objects a simple blob detector can find).

Label semantics differ from the chest domain. CBIS-DDSM is a curated set of lesion cases with no normal mammograms at all, so the binary label is malignant vs. benign, not healthy vs. diseased; the two edit directions are benign→malignant and malignant→benign, and a successful malignant→benign edit claims only that the lesion was made to *look* benign to the classifier, not that disease was removed.

Table 14: Mammography-domain evaluation, $n_{\text{eval}} \leq 200$ per group (`checkpoints/cbis_ddsm_breast/eval_metrics.json`; transfer rows from `.../heldout_transfer.json`). The CV row and the held-out transfer row are deliberately in the same table: CV is computed against the classifier that guided generation and means nothing on its own.

Metric	Definition	ben.→mal.	mal.→ben.
CV	fraction valid ($\tau = 0.8$), *against the guidance model*	1.000	1.000
	mean	1.000	
Transfer	**independent validators, mean $\Delta_{\text{val}}/\Delta_{\text{guid}}$**	**0.066**	**0.056**
	resnet18-heldout (AUC 0.723)	0.116	0.032
	effnet-b0-heldout (AUC 0.693)	0.017	0.080
	same-corpus validator (AUC 0.988†)	*0.476*	*0.327*
	CMMD external (AUC 0.507/0.499 — uninterpretable)	*0.091 / −0.006*	*0.052 / 0.095*
FID	breast-domain features, lower better	163.6	193.8
IP	MS-SSIM / HF-corr	0.891 / 0.932	0.886 / 0.946
EditConcentration	top-5% mass (uniform = 0.05) / % pixels changed	0.367 / 38.2%	0.332 / 40.6%
MOP	marker preservation	0.930	0.872
LLR Δ	(t, p)	+0.00014 (0.04, 0.970)	−0.00107 (−0.34, 0.732)
MSI Δ	(t, p)	−0.082 (−1.39, 0.165)	−0.055 (−1.68, 0.094)
BDC Δ	(t, p)	+0.0014 (0.29, 0.773)	+0.0154 (2.46, 0.015*)

† `resnet18-breast-cbis`, the backbone the domain profile reserves as its held-out validator, was trained on the same patient split as the guidance model and therefore on 336 of these 400 eval images (84%). Its AUC of 0.988 measures memorisation, and its transfer ratio is an upper bound, not an independence check — which is why it is excluded from the mean.

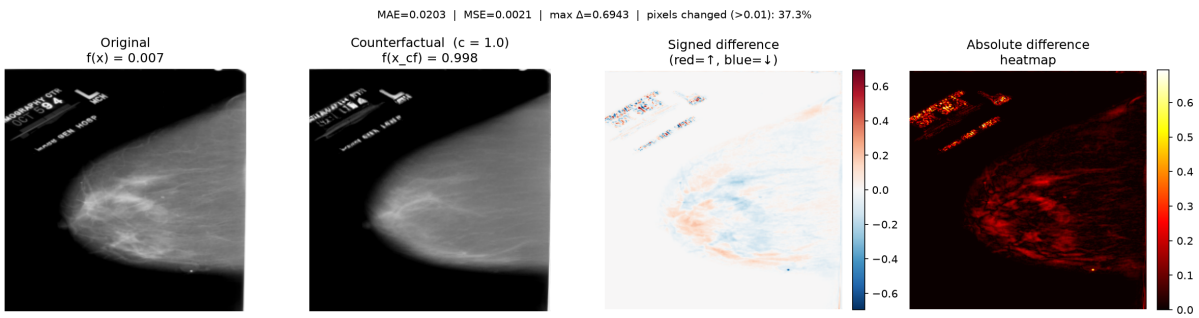


Figure 7: **Benign → malignant**, mammography domain. Same four-panel layout as Figures 1–2.

Table 15: Mammography held-out transfer ratio per validator and direction, $n = 200$ per group, percentile bootstrap over 10,000 resamples with numerator and denominator resampled jointly to preserve within-image pairing. Verdict thresholds are `arm_transfer.py`'s: < 0.15 adversarial, < 0.5 partial. AUC is measured on the 400 real originals (190 benign / 210 malignant) against radiologist pathology labels; ρ is rank agreement with the guidance model on those same unedited images.

Validator	Independence	AUC	ρ	ben.→mal. (95% CI)	mal.→ben. (95% CI)
Independent and competent — these are the result					
resnet18-heldout	patient-disjoint	0.723	+0.385	0.116 [0.097, 0.136]	0.032 [0.017, 0.047]
effnet-b0-heldout	patient-disjoint	0.693	+0.370	0.017 [−0.011, 0.044]	0.080 [0.057, 0.103]
mean				**0.066**	**0.056**
Reported for contrast — not independence checks					
resnet18-cbis	same train fold	0.988	+0.826	0.476 [0.427, 0.526]	0.327 [0.282, 0.373]
resnet18-cmmd	external corpus	0.507	+0.055	0.091 [0.071, 0.111]	0.052 [0.034, 0.069]
effnet-b0-cmmd	external corpus	0.499	−0.000	−0.006 [−0.019, 0.007]	0.095 [0.084, 0.107]

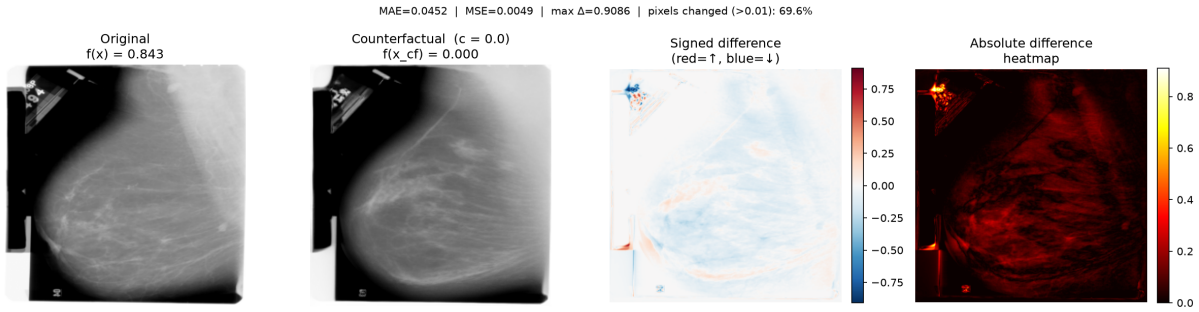


Figure 8: **Malignant → benign**, mammography domain.

This is not a second validated success, and should not be read as one. A mean CV of 1.000 against the guidance classifier looks, on its face, like an even stronger number than the RSNA result. It is not a result at all: it is measured against the model whose gradient produced the edits. The held-out re-scoring reported alongside it in Table 14 is what that number has to be read next to, and the two disagree by an order of magnitude.

Verdict: adversarial, in both directions, distinguishable from noise. Against validators that never saw the guidance model's training data and never saw a single one of the 400 images they are scoring, the counterfactuals reproduce a mean 6.6% (benign→malignant) and 5.6% (malignant→benign) of the guidance model's own score shift. Every one of the four bootstrap intervals in Table 15 lies entirely below the 0.15 adversarial threshold — the largest upper bound is 0.136 — so this is not a small sample failing to resolve a real effect; at $n = 200$ per group the data positively establishes the adversarial regime. In the benign→malignant direction the EfficientNet validator's interval $[-0.011, 0.044]$ includes zero outright, meaning that for that validator the edits carry no detectable malignancy signal whatsoever. For comparison, the chest domain's worst arm transfers at $0.230/0.146$ and its committed default at $0.282/0.293$: mammography is markedly *more* adversarial than any RSNA arm, not less.

The reserved validator was never an independence check, and the gap quantifies what that cost. `resnet18-breast-cbis`, which `domains/cbis_ddsm_breast.py` designates as this domain's held-out validator, was trained on the same patient split (`seed` 0, `val_frac` 0.2) as the guidance model, and therefore on 336 of the 400 eval images — 84%, across 304 of 362 patients. Its AUC of 0.988 on those images is memorisation, against 0.723 and 0.693 for validators that have not seen them. Had its transfer ratio been reported as the domain's independence check, it would have read $0.476/0.327$ — "partial" on this project's own scale, and between four and ten times the honest figure. The anti-circularity check in `DomainProfile.validate()` passes here, because it compares

model *identity*; nothing in the pipeline was comparing training *data*, and that is the gap this measurement closes.

An external corpus was tried and did not survive its own sanity check — and both its in-domain and cross-domain AUC are reported, not just the failing one. A validator trained on CMMD — a different country, a different scanner generation, no possible patient overlap — would in principle have been the strongest available independence claim. On its *own* held-out CMMD patients it reaches out-of-fold AUC 0.624 (ResNet-18) / 0.635 (EfficientNet-B0): above chance, but already below this project's 0.65 floor before CBIS-DDSM is involved at all. Scored on real CBIS-DDSM images it drops further, to 0.507 / 0.499 — indistinguishable from chance — with rank agreement $\rho = +0.055$ and $\rho = -0.000$ against the guidance model on the unedited originals. The scores are not degenerate (std 0.108, unimodal over $[0.18, 0.82]$), and label mapping and pixel preprocessing are byte-identical to CBIS-DDSM's; what differs is framing — CMMD's mirror ships pre-cropped lesion patches (34.5% near-black background) against CBIS-DDSM's full mammograms (54.7%) — consistent with the classifier keying on something framing-specific rather than on transferable malignancy signal. This was a mediocre validator on its own data before the domain shift compounded it. Its transfer ratios (0.091, -0.006) are therefore reported as *uninterpretable* and excluded from the mean, because a classifier that cannot separate real malignant from real benign would not follow a genuine disease edit either — a low ratio from it would be evidence about the validator, not about the counterfactuals. That distinction is the reason every validator here is gated on AUC before its ratio is read.

Two independent lines of evidence now agree. The clinical proxy metrics reached this conclusion first and without a classifier: in the malignant→benign direction BDC shifts significantly ($p = 0.015$) while LLR — the metric that would move if a lesion actually shrank — does not ($p = 0.732$), and edit concentration (0.33–0.37 against a uniform baseline of 0.05, with 38–41% of all pixels changed) is far short of a focal, lesion-confined edit. The held-out transfer ratio is a second, methodologically unrelated instrument — an independently trained classifier rather than a segmentation-derived statistic — and it points the same way. The convergence matters more than either result alone: a diffuse density wash that moves a density-sensitive classifier without altering the lesion is precisely an edit that a classifier trained on different patients would not follow, and that is exactly what is observed.

Two caveats keep this honest in the other direction. The independent validators are competent but not strong (AUC 0.69–0.72), so they set a floor on how much real signal an edit must carry to register, and a subtle genuine lesion change could fall below it; the ratio is evidence of a large adversarial component, not proof that the signal is exactly zero. And BDC's $p = 0.015$ does not survive Bonferroni correction across the six clinical tests ($0.05/6 = 0.0083$), a caveat that applies to the clinical-metric line of evidence and not to the transfer ratio, which is one pre-specified measurement rather than a family of six. Neither caveat disturbs the main finding, but the exact scope of "disjoint" does need stating. These validators are disjoint from the guidance model's *training* fold and from every image they score. They are trained on its *validation* fold — CBIS-DDSM splits 1,168 / 292 patients, so excluding both folds would leave zero to train on — and they are not disjoint from the PPO rollout pool, which no subset of this corpus can be, since the far-pool scan drew on all 2,857 images (1,603 plus 1,254, exactly the corpus). The validation-fold overlap biases *against* the reported result: a validator sharing images with the guidance checkpoint's selection criterion should agree with it more, not less, so $0.066/0.056$ is a conservative figure.

What this extension demonstrates, then, is stronger than the architecture-generalises claim it was originally written to support. The domain-registry abstraction did carry to a modality sharing nothing with chest radiography. But the substantive finding is that a counterfactual system reporting perfect validity against its own guidance classifier reproduced, on a second modality, the same adversarial shortcut that held-out validation exposed on the first — and that the check which was supposed to catch it had been quietly disabled by a shared training split rather than by a shared model.

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